

Resolution Tier (0–15) Boundary Check & Spatial Monad Integrity in the Web of Life Simulation Engine

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Sprint 028 Technical Report

Abstract

Modeling complex biospheres computationally requires strict adherence to spatial and thermodynamic conservation laws. The Web of Life simulation engine utilizes the Uber H3 hierarchical hexagonal grid to simulate global biogeochemical cycles and trophic energy diffusion. Sprint 028 introduces formal resolution tier boundary validation functions (`isValidResolution` and `assertValidResolution`) that enforce strict integer bounds within the permissible range $[0, 15]$. By embedding these validation guards into spatial monad state transitions, the engine prevents spatial leakage, ensuring strict compliance with mass conservation (First Law of Thermodynamics) and accurate multi-scale energy aggregation.

1 Introduction and Thermodynamic Context

The Web of Life simulation models Earth’s biosphere through discrete spatial monads (`SpatialMonad`) layered over an H3 hexagonal grid. Thermodynamic consistency—specifically the conservation of matter and energy across spatial scales—depends entirely upon valid hierarchical indexing.

Unbounded or fractional resolution tiers can corrupt parent-child transformations (`h3ToParent` and `h3ToChildren`), leading to artificial matter creation or destruction. Sprint 028 establishes rigorous boundary checks within `src/spatial/h3_grid.ts` to safeguard the simulation against illegal state spaces.

2 Mathematical Formalization of Spatial Bounds

Let $\mathcal{R}$ be the set of permissible H3 resolution tiers:

$$\mathcal{R} = \{r \in \mathbb{Z} \mid 0 \leq r \leq 15\}$$

For any spatial monad operation M_r operating at resolution r , the validation predicate $\psi(r)$ is defined as:

$$\psi(r) = \begin{cases} 1 & \text{if } r \in \mathcal{R} \\ 0 & \text{otherwise} \end{cases}$$

If $\psi(r) = 0$, execution halts via a `RangeError`, preventing illegal state transitions that violate closed-system thermodynamic conservation equations. Across valid hierarchical transitions, mass (ΔM) and energy (ΔE) satisfy:

$$\sum_{i=1}^7 M_{\text{child},i} = M_{\text{parent}}$$
$$\sum_{i=1}^7 E_{\text{child},i} = E_{\text{parent}} - E_{\text{dissipation}}$$

3 Implementation Contract

The implementation within `src/spatial/h3_grid.ts` is defined as follows:

```
export function isValidResolution(resolution: number): boolean {
  return Number.isInteger(resolution) && resolution >= 0 && resolution
    <= 15;
}

export function assertValidResolution(resolution: number): void {
  if (!isValidResolution(resolution)) {
    throw new RangeError(`Invalid H3 resolution tier: ${resolution}.
      Must be an integer between 0 and 15.`);
  }
}
```

4 Conclusion

Sprint 028 successfully fortifies the spatial indexing subsystem of the Web of Life simulation engine. By enforcing tier boundaries $[0, 15]$, the engine maintains strict thermodynamic monad integrity across all trophic energy diffusion and spatial aggregation processes.

Repository Reference: <https://github.com/pascalranoroarijaona/WebOfLife>